

MoveitScene Design

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1. Usage Environment

Motherboard: Ubuntu 22.04

ROS2: Humble

2. Driving the Real Machine

Driving the real machine involves subscribing to Moveit2's `/joint_states` topic to convert the joint state information of the robotic arm into control data for the actual robotic arm.

Note: Since the real robotic arm lacks obstacle avoidance, some poses may encounter obstacles; therefore, the planned robotic arm movements should be as reasonable as possible and avoid obstacle-prone areas.

(It is recommended to use preset poses for the demonstration of driving the real machine)

2.1 Starting the Real Machine

Without driving the real machine, demonstrate the robotic arm movements in MoveIt simulation:

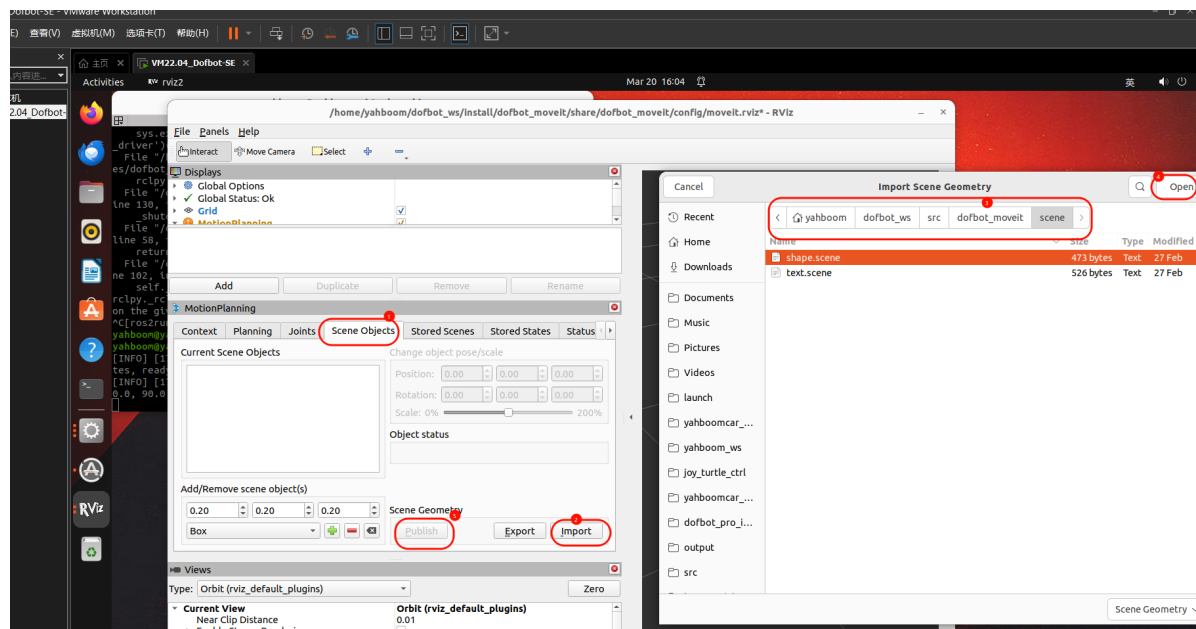
```
ros2 run dofbot_driver dofbot_driver
```

2.2 Starting MoveIt2

```
ros2 launch dofbot_moveit demo.launch.py
```

3 Scene Design

Add a scene in RViz2



Then click the publish option.

Startup Command

The robotic arm needs to load successfully in MoveIt and display `You can start planning`. Now! Run the following command:

```
ros2 run dofbot_moveit set_scene
```

